



RAMAIAH
UNIVERSITY
OF APPLIED SCIENCES

Final Presentation

Project Work-II
B-TECH, ECE
Batch 2022

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Title of the Project

Autonomous UAV System for Real-Time Human Detection and Health Vitals Monitoring

- **Supervisor : Mr. Viswanath K**
- **Place of Work : RTC, Peenya**

Outline

- Title Slide
- Introduction
- Objectives
- Contactless Medical Triage
- Software Pipeline
- Telemetry-Guided Noise Cancellation
- Empathetic Human-Robot Interaction
- Smart Drop Kits
- Gantt chart
- Conclusion
- SDG
- Reference's

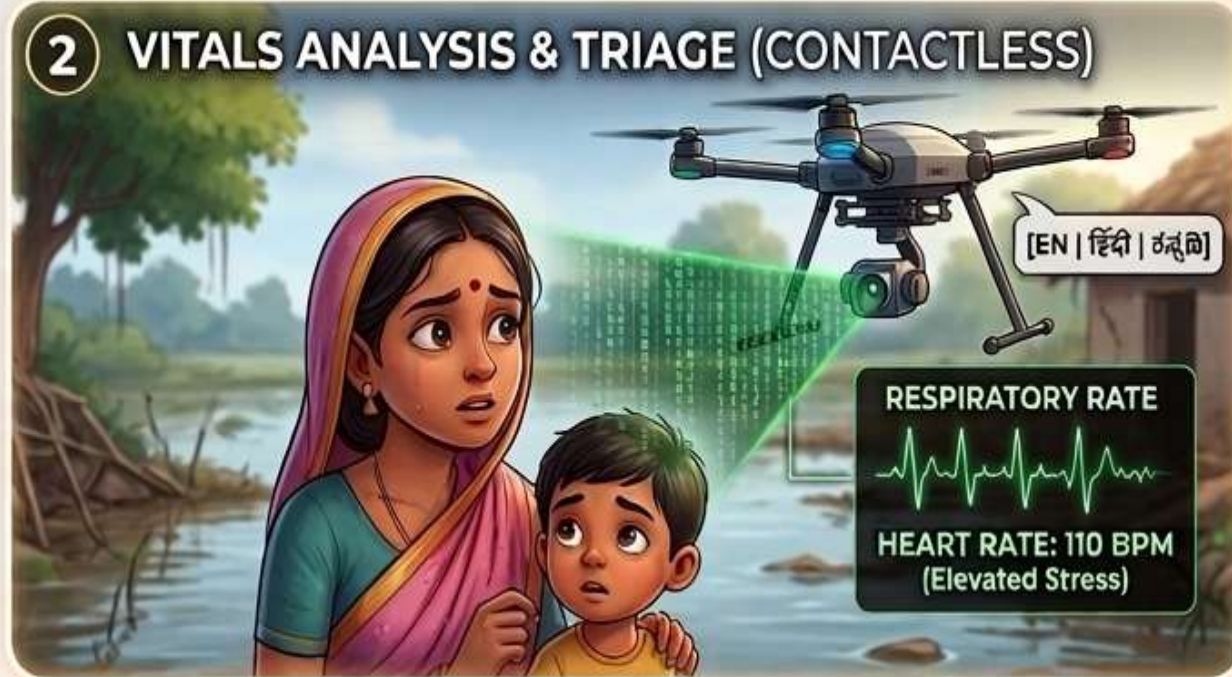
Introduction

- **Background:** Natural disasters and industrial accidents require rapid, accurate Search and Rescue (SAR) deployments to maximize the chances of survival.
- **Our Foundation (7th Semester):** We successfully developed and demonstrated a manual UAV capable of visual pose estimation (identifying "Hands Up" as a distress signal).
- **The Evolution (8th Semester):** Autonomous Drone capable of AI2P interaction and human vitals detection
- **The Core Concept:** using Computer Vision, Edge AI, to create a drone that can see, hear, and interact with victims in denied environments.

1 SCAN & APPROACH (AI DETECTION)



2 VITALS ANALYSIS & TRIAGE (CONTACTLESS)



3 INTERACTION & DROPPING KITS (MULTILINGUAL)



4 DIGITAL BRIDGE & COORDINATION (GROUND TEAM)



Title : Autonomous UAV System for Real-Time Human Detection and Health Vitals Monitoring

Aim: To Design and Develop an empathetic interaction Autonomous Drone with AI, remote Photoplethysmography using image processing for vital detection of human being

Objectives

1. To carry out literature and review for contactless Heart rate monitoring, Making Drone Autonomous.
2. To design and develop an autonomous flying hexacopter platform with Pixhawk and mission planner.
3. To Implement image processing techniques and AIML algorithms on the Pi 5 to extract Heart Rate and Respiratory Rate via camera feed.
4. To implement autonomous human detection, target locking and victim tracking algorithms that enable the drone to automatically approach and continuously monitor a detected person.
5. To implement and develop an android app for visual outputs and information then Test and validate the system.

Methodology for contactless vital detection

Concept: Contactless vital monitor, using a standard camera to replicate a medical device

Heart Rate (PPG): The system detects micro-fluctuations in skin color that are invisible to the human eye but correspond to blood flow from the heart.

Respiratory Rate (Kinematics): The system tracks the physical rise and fall of facial landmarks(Nose) to calculate breaths per minute.

Region of Interest (ROI): a 468-point 3D Face Mesh to lock onto the forehead for heart rate and the nose bridge for breathing, ensuring the data remains stable even if the victim moves

rPPG: Every time your heart beats, it pumps a fresh wave of blood into your face. This causes the tiny blood vessels in your forehead to expand and fill up for a fraction of a second.

Because blood contains hemoglobin, it absorbs green light. So, when a fresh wave of blood rushes into your face with every heartbeat, your skin microscopically absorbs a tiny bit more green light from the room, making the skin reflection a tiny bit darker. When the blood flows back out, the skin gets a tiny bit brighter.

This color change is completely invisible to the human eye. But to a computer reading a camera, it shows up as a shifting wave of numbers. This technique is called **rPPG (Remote Photoplethysmography).**"

Hardware-to-Software Data Bridge

- **Bayer Filter Mosaic:** Camera sensor captures raw light intensity voltages in a checkerboard grid.
- **Demosaicing Interpolation:** OpenCV reconstructs missing color data by averaging neighboring pixels.
- **3D Matrix Construction:** Compiles data into a 3D NumPy Array organized in **BGR sequence**.
- **Target Extraction:** Code isolates **Channel Index 1** (Green) as our main physiological carrier wave.

Signal Extraction

Step 1: Temporal Normalization

The Equation:

$$C_n(t) = C(t) / \mu C$$

Where **C(t)** is the raw RGB vector at a specific frame, and **μC** is the average RGB value over that entire time window.

Official Name: Spatio-Temporal DC-Normalization

The Concept:

99.5% Constant baseline skin color and room lighting.

0.5% Shifting cardiac pulse wave.

The Impact: Deletes the massive constant baseline, ensuring distance invariance and handling skin color variation.

Steps 2 & 3: Noise Cancellation

The Equations:

$$X(t) = CnR(t) - CnG(t)$$
$$Y(t) = Cn_{R(t)} + Cn_{G(t)} - (2 * Cn_{B(t)})$$

Where ***CnR*** ***CnG*** , and ***CnB*** are the normalized Red, Green, and Blue channels. X(t) and Y(t) represent a space where uniform intensity variations along the [1, 1, 1] axis have been completely canceled out.

Official Names: Orthogonal Subspace Projection & Adaptive Subtraction via Alpha Tuning

The Concept: Drone shaking affects all color channels identically at the same millisecond.

The Impact: Perpendicular coordinates force vibration noise to balance to **zero**. Alpha acts as an automatic volume knob to subtract remaining surface glare.

The Equations:

$$\alpha = \text{std}(X) / \text{std}(Y)$$

The raw pulse segment $h(t)$ for the current window is computed as:

$$h(t) = X(t) - (\alpha * Y(t))$$

Where **std(X)** and **std(Y)** represent the standard deviation of the X and Y signals over the window, and **h(t)** is the extracted raw pulse segment.

Official Engineering Name: Adaptive Signal Scaling and Dynamic Subtraction.

The Core Concept: * $Y(t)$ represents the isolated motion and lighting noise axis.

$X(t)$ contains the target heartbeat signal mixed with residual glare.

The Functional Impact:

The Volume Knob: Alpha dynamically calculates the exact intensity of the noise by comparing the standard deviations of the two channels.

Destructive Interference: Multiplying alpha by $Y(t)$ and subtracting it from $X(t)$ perfectly cancels out leftover surface distortions (like sweat glare or passing shadows).

The Output: Isolates the pure, uncorrupted time-domain cardiovascular pulse wave $h(t)$.

Step 4: Signal Reconstruction

The Equation:

$$P(t) = P(t) + h(t) - \mu h$$

Official Name: Overlap-Add (OLA) Signal Reconstruction

The Concept: * Heavy matrix math is split into short, sliding 2-second windows to save memory on the Pi 5 hardware.

The Impact: Acts as digital tape. Subtracts the window mean μh and stitches segments together into a smooth, continuous wave $P(t)$.

Software Stack

MediaPipe: 3D face mesh model - find and track the face in 3D space.

OpenCV: used to draw the HUD and process the video frames.

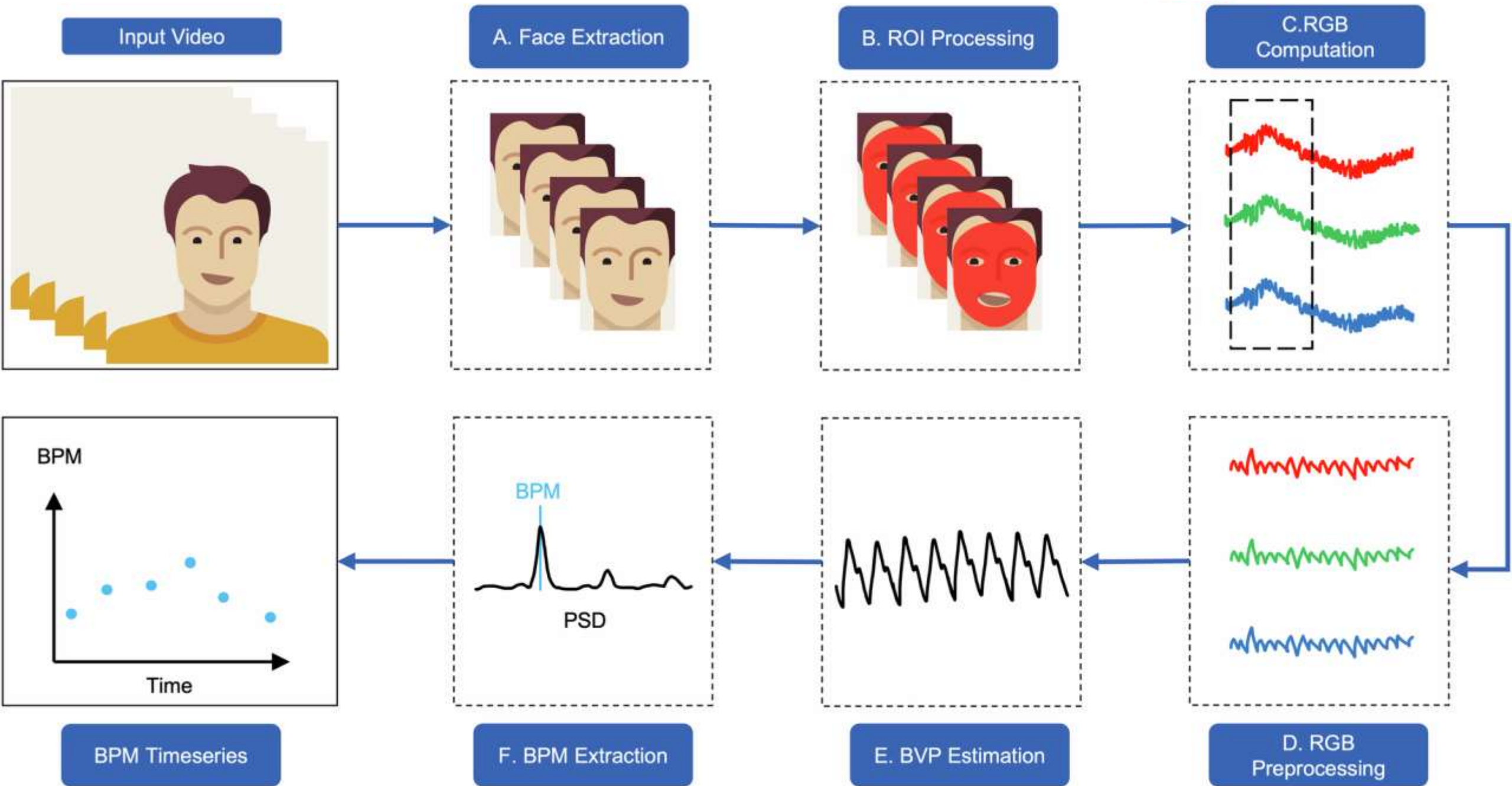
SciPy/NumPy: used to run the heavy mathematical filters and signal analysis.

4th Order Butterworth Bandpass Filter: For Heart Rate, it only allows frequencies between 0.75 Hz and 3 Hz (45–180 BPM). to block out "non-human" frequencies high-frequency drone rotor vibrations

Welch's Power Spectral Density: Instead of just looking at a messy wave, Welch's method breaks the signal into segments and averages them. reduces "noise spikes" and allows the system to identify the true, dominant heart and breath rates with much higher stability than a standard Fast Fourier Transform (FFT)

Signal Quality Index - SQI: A real-time self-assessment tool that compares the strength of the "Pulse" against the background "Noise." If the SQI is low, the HUD turns yellow/red to warn the operator that the data is currently unreliable due to movement or lighting.

Non-Linear Median Filter : A final decision-making layer that looks at a window of the last 20 readings. If the system suddenly detects a fake jump to 200 BPM for one frame (a glitch), the Median filter discards it, keeping the display smooth and consistent



This analysis examines the heart rate (BPM) response during three minutes by isolating specific states Active, Peak, and Recovery the data was collected under varying environmental and physical conditions that typically effect the ppg such as **Against Light, With Light, Light & Sweat**. we can quantify the efficiency of the system in transitioning from moderate activity to high-intensity stress and back to a resting state.

The three phases are defined as follows:

Phase 1: Active (0:00 – 1:00 min) Initial readings taken during steady-state movement or a warm-up. This serves as the **baseline** to determine the starting intensity before peak exertion.

Phase 2: Peak Workout (1:01 – 2:00 min) Readings captured during maximum physical demand. This phase identifies the **Peak Heart Rate**, reflecting the body's ability to handle high intensity stress.

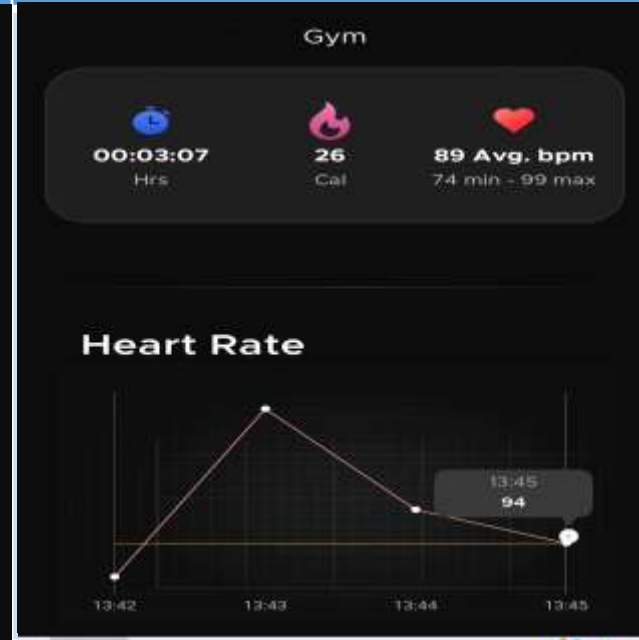
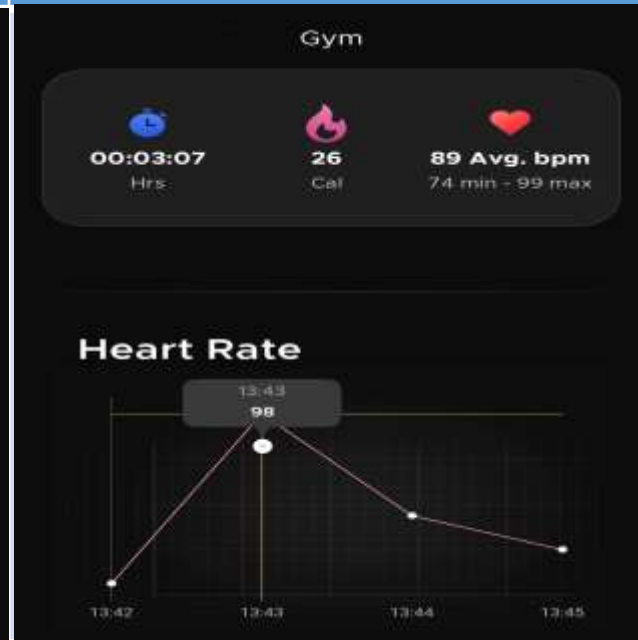
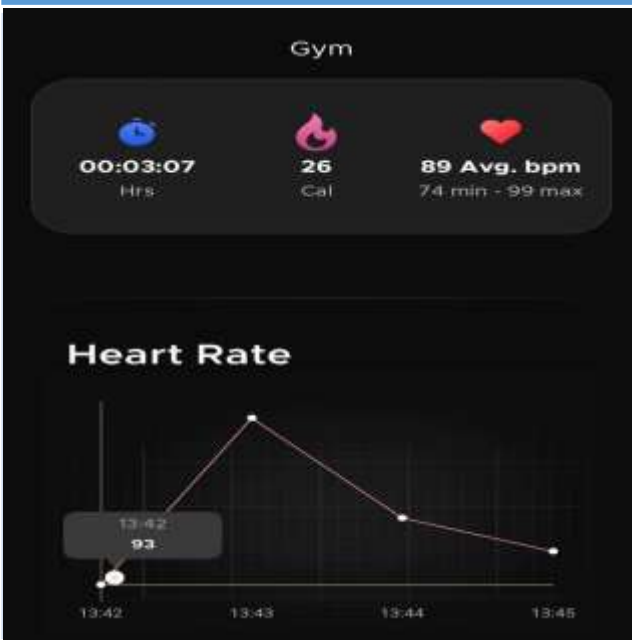
Phase 3: Recovery (2:01 – 3:00 min) Readings taken immediately following the peak phase. This is the most critical metric for assessing fitness, as it calculates the **Heart Rate Recovery (HRR)**—the speed at which the heart returns to baseline.

Against Light Active (1st min)

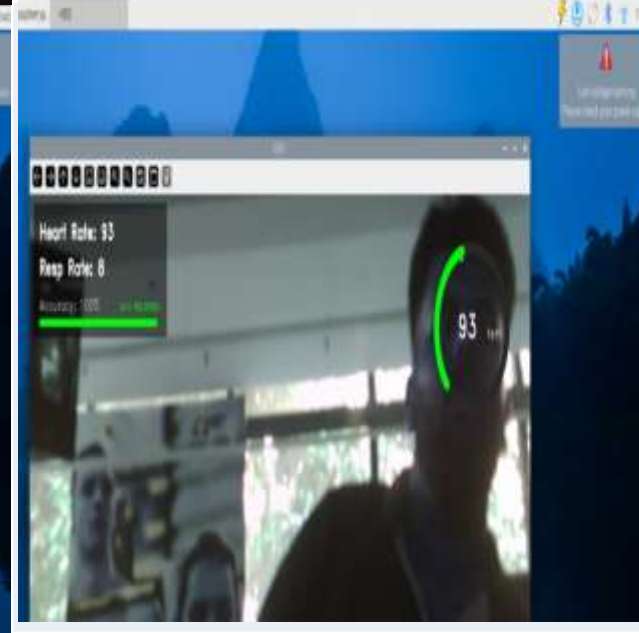
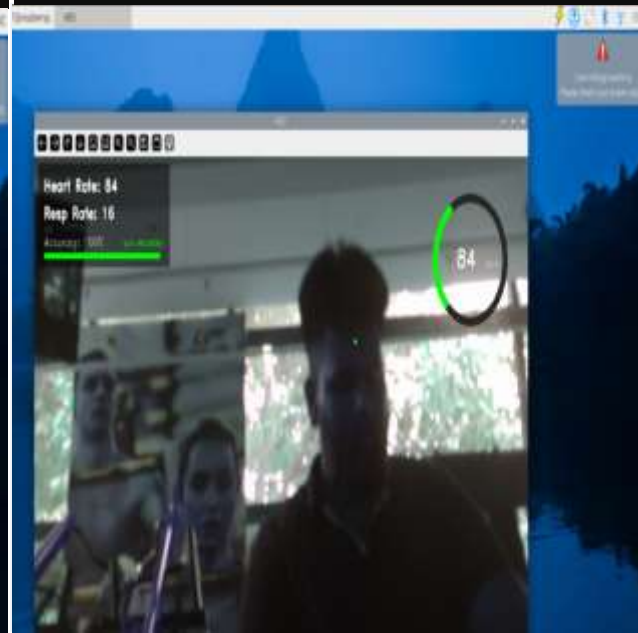
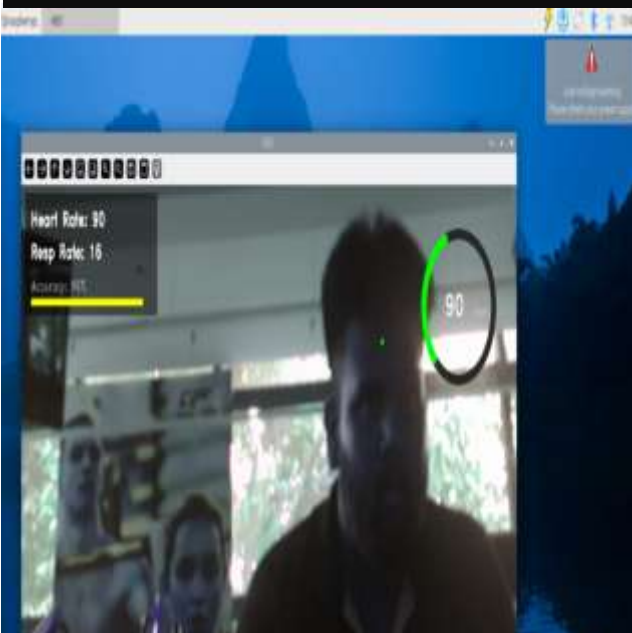
Peak (2nd min)

Recovery (3rd min)

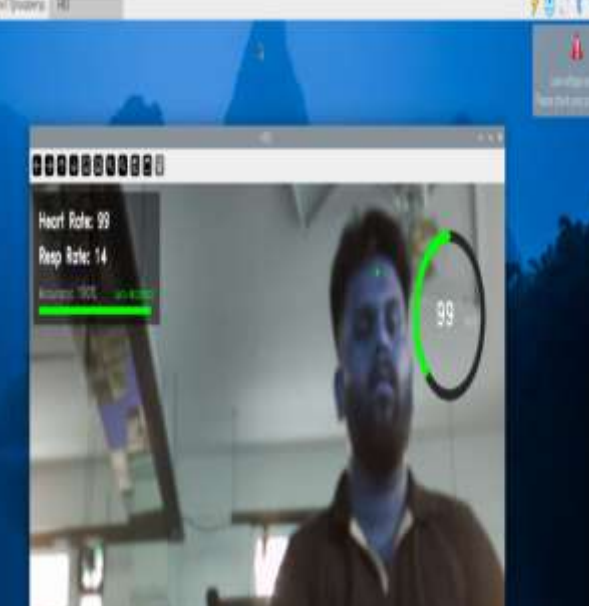
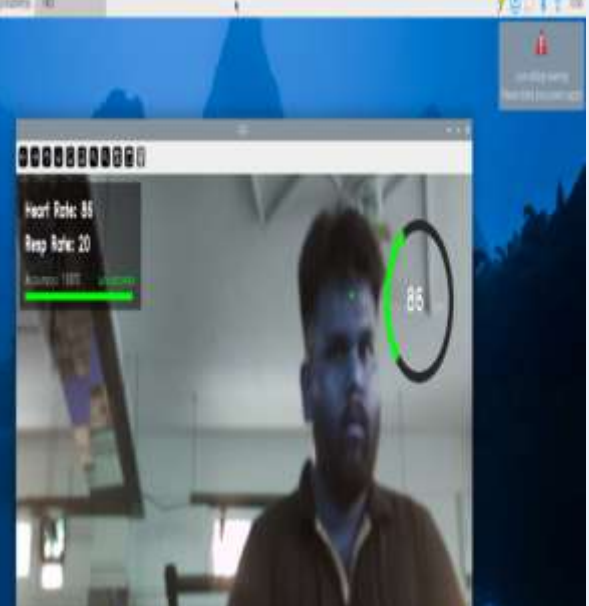
Results



- Avg bpm = 89
- Reference here is the heart rate average/minute



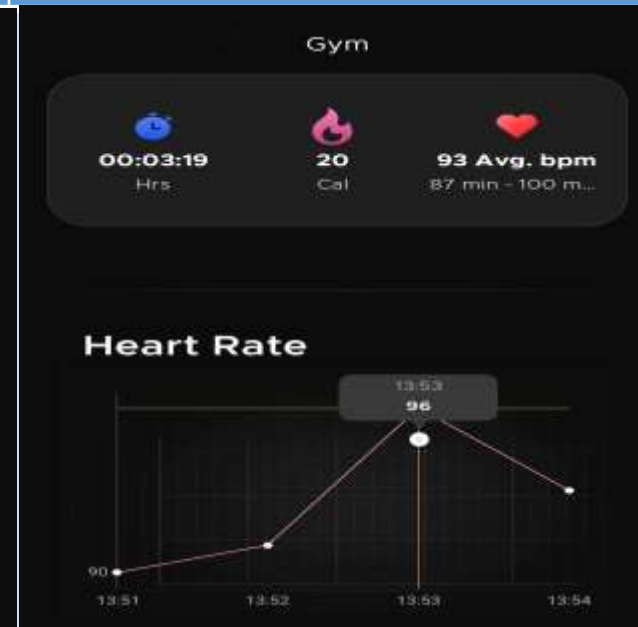
- The output is the closest reading to the reference.
- Peak heart rate readings are noted in a min
- Some instances showed that the face was undetectable

With Light Active (1 st min)	Peak (2 nd min)	Recovery (3 rd min)	Results
			- Avg bpm = 92 - Reference here is the heart rate average/minute
			- The output is the closest reading to the reference. - Peak heart rate readings are noted in min - The face was detected even with rapid hand movements near the head

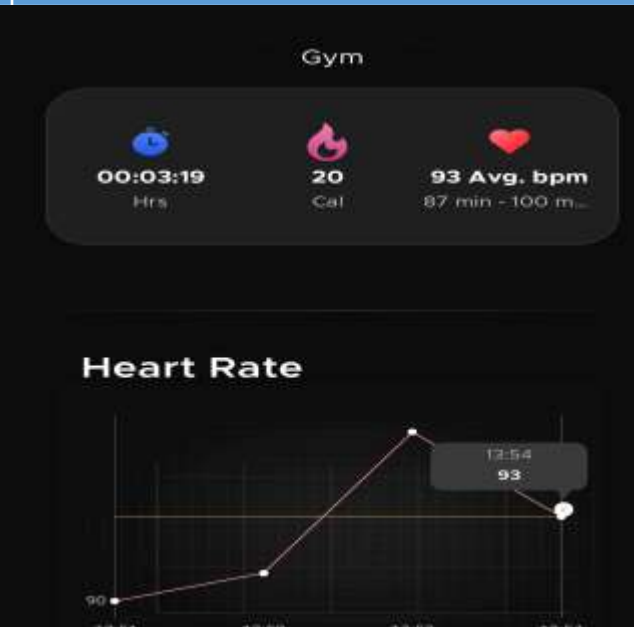
With Light and sweat Active (1st min)



Peak (2nd min)

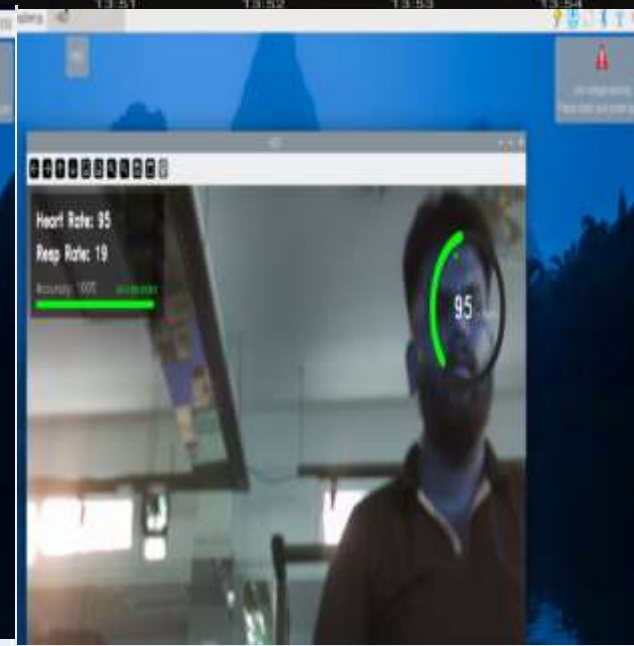
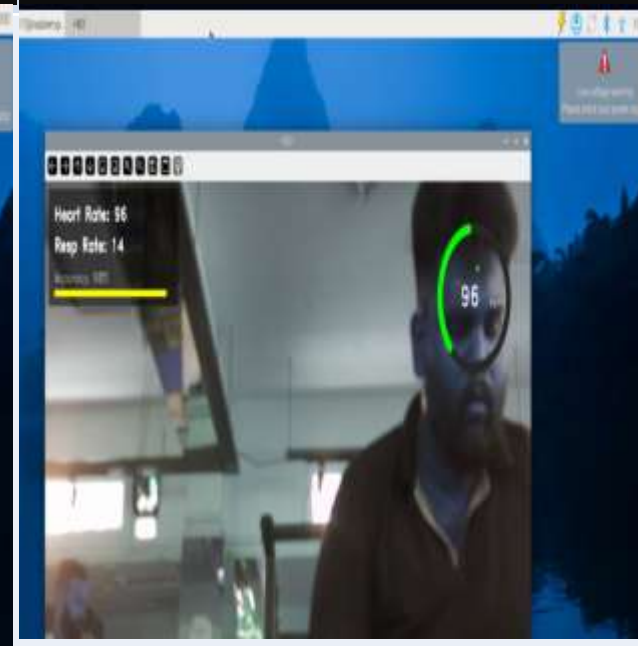
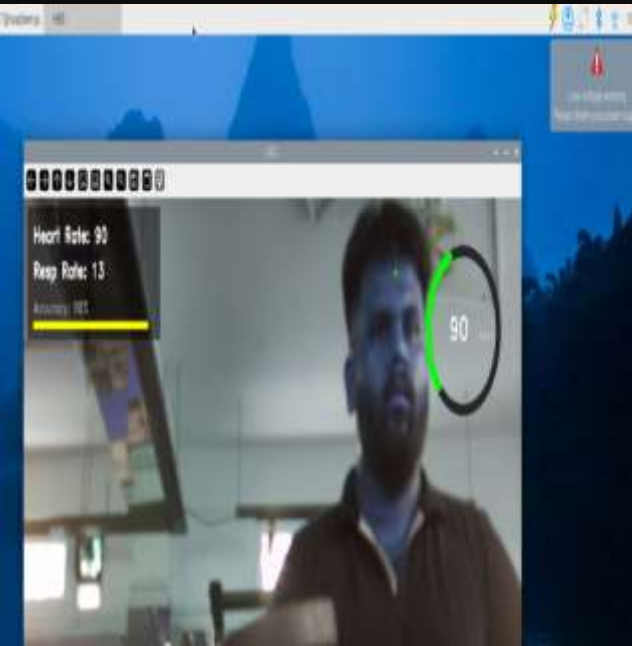


Recovery (3rd min)



Results

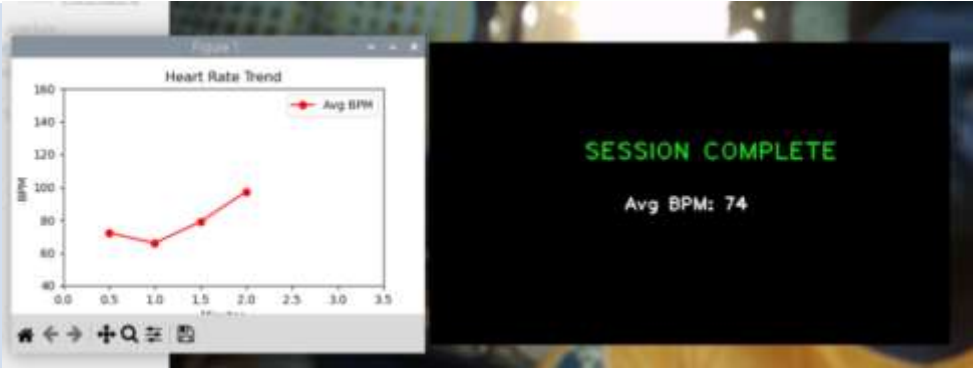
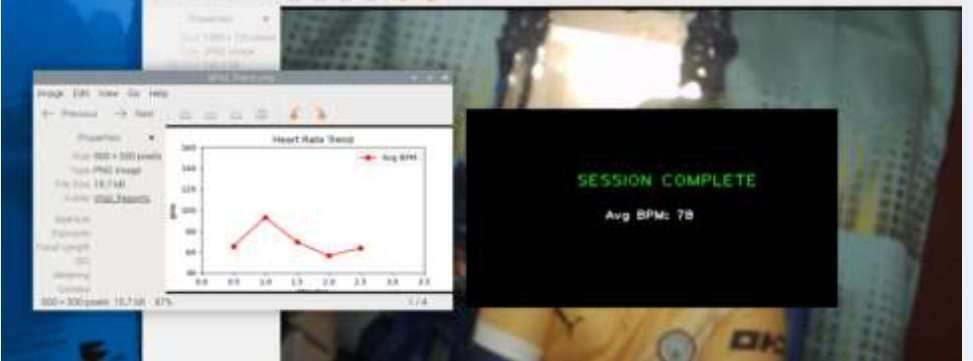
- Avg bpm = 93
- Reference here is the heart rate average/minute



- The output is the closest reading to the reference.
- Peak heart rate readings are noted in min
- The face was detected even with rapid hand movements near the head and sweat on the forehead

Against Light	1:43 pm 96 bpm	79, 73, 90, 88, 51, 80, 118, 125, 136, 51, 58, 63, 64, 73, 73, 84, 98 Avg = 82.59 bpm	96-82.59 bpm = 15.41 units lower Percent Error = 15.73%	Accuracy = 100-15.73 Accuracy = 84.27%
With Light	1:48 pm 92 bpm	100, 110, 106, 64, 72, 69, 89, 84, 89, 83, 90, 87, 90, 91, 85, 78, 77 Avg = 86.12 bpm	92-86.12 bpm = 5.88 units lower Percentage Error = 6.39%	Accuracy = 100-6.39 Accuracy = 93.61%
Light and Sweat	1:53 pm 96 bpm	63, 88, 54, 60, 71, 72, 79, 86, 85, 88, 91, 89, 83, 85, 74, 87, 96,111 Avg = 81.22 BPM	96-81.22 bpm = 14.78 units lower Percentage Error = 15.39%	Accuracy = 100-15.39 Accuracy = 84.61%

Distance Accuracy Measurement

Distance	Smart watch (3min reading)	Output heart rate (3 min reading)	Accuracy
1 meter			91.35%
2 meter			91.76%

Accuracy of system

Test Variable	Smart Watch	System Output (BPM)	Error (%)	Accuracy (%)
Distance: 1 Meter	81 BPM	74 BPM	8.65 %	91.35%
Distance: 2 Meters	85 BPM	78 BPM	8.24 %	91.76%
Against Light	96 BPM	82.59 BPM	15.73 %	84.27%
Direct Light	92 BPM	86.12 BPM	6.39 %	93.61%
Light + Sweat	96 BPM	81.22 BPM	15.39 %	84.60%

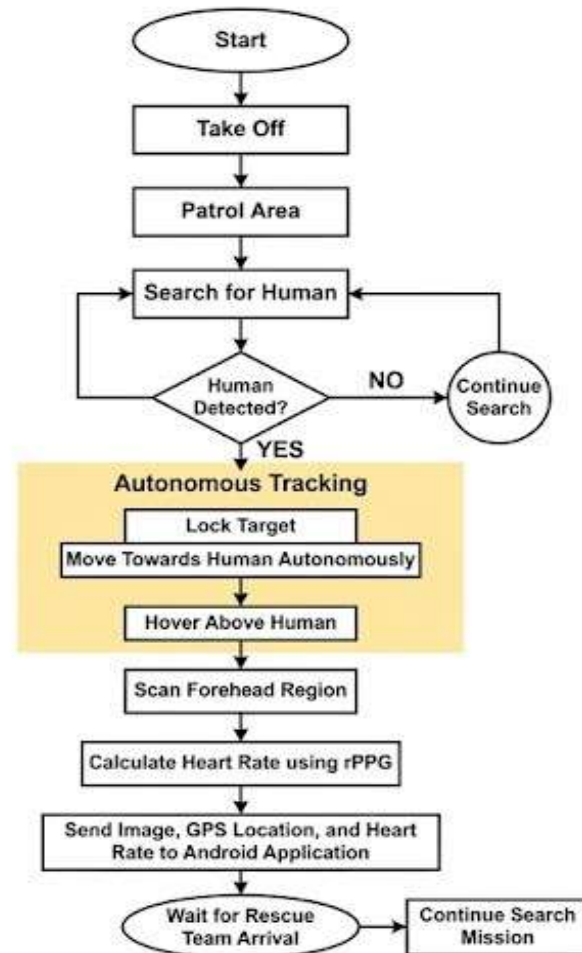
The system successfully validates the use of contactless rPPG for emergency response, maintaining over 91% accuracy across varying distances and lighting conditions

Autonomous Navigation via MAVLink Telemetry

- **Concept:** The Raspberry Pi communicates with Pixhawk through MAVLink.
- **Human Detection:** The onboard camera continuously scans the environment.
- **Target Locking:** Once a human is detected, the target is locked.
- **Autonomous Approach:** The drone automatically moves toward the detected person.
- **Hover Mode:** The drone maintains a stable position near the victim.
- **Vitals Monitoring:** Heart-rate monitoring starts automatically after target acquisition.

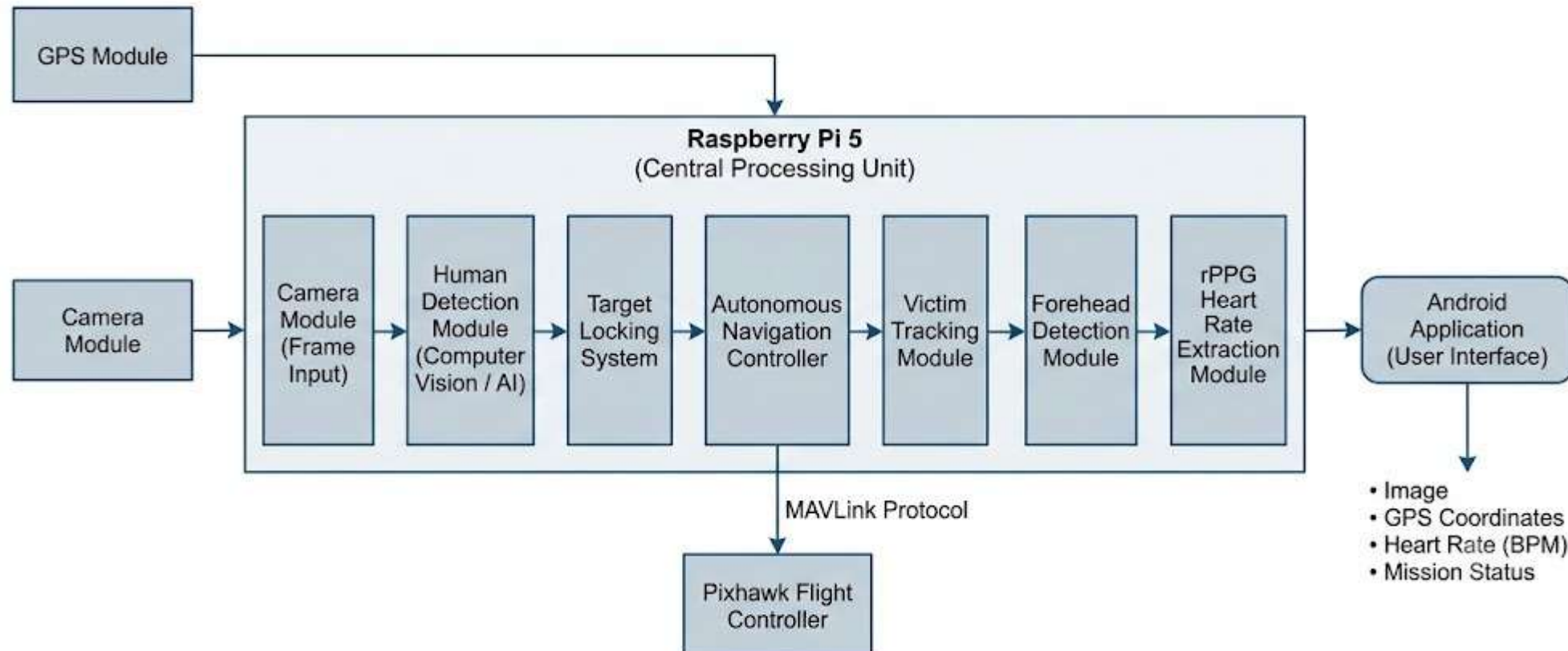
Flow Chart:

Autonomous Human Detection and Vitals Monitoring Workflow



System Architecture:

Autonomous Human Detection and Contactless Vitals Monitoring Drone Architecture



Tools & Libraries

OpenCV:

- Camera frame processing
- Human detection & tracking

YOLO

- Real-time human detection
- Target localization

NumPy & SciPy

- Numerical computation
- Signal processing for rPPG

MediaPipe Face Mesh

- Facial landmark detection
- Forehead ROI extraction

PyMAVLink

- Raspberry Pi ↔ Pixhawk communication
- Autonomous navigation commands

Raspberry Pi 5

- AI processing and decision making

Pixhawk

- Flight control and autonomous movement

APP Frontend Libraries & Packages

Technology

- Flutter SDK
- Dart Programming Language

➤ **web_socket_channel**

Used for real-time communication
Connects app to backend via WebSocket
Receives live JSON data

➤ **provider**

Used for state management
Stores and manages incoming data
Updates UI automatically

➤ **fl_chart**

Used to display live charts
Shows Heart Rate, Respiratory Rate

➤ **flutter_map**

Used for map visualisation
Displays live drone location

➤ **latlong2**

Supports map functionality
Handles latitude and longitude

➤ **shared_preferences**

Used for local storage
Saves IP address and connection settings

➤ **Built-in Libraries**

dart: convert → JSON parsing
dart: async → async operations

APP Backend Libraries & Packages

Technology

- Python Programming Language

➤ **websockets**

Creates a WebSocket server

Sends real-time data to the app

➤ **opencv-python**

Used for image processing

Captures and processes camera frames

➤ **numpy**

Used for numerical operations

Handles signal data

➤ **scipy**

Used for signal processing

Performs filtering and FFT

➤ **pymavlink**

Used for communication with drone

Reads GPS, battery, altitude

➤ **Built-in Libraries**

asyncio → async server handling

json → data formatting

serial → hardware communication

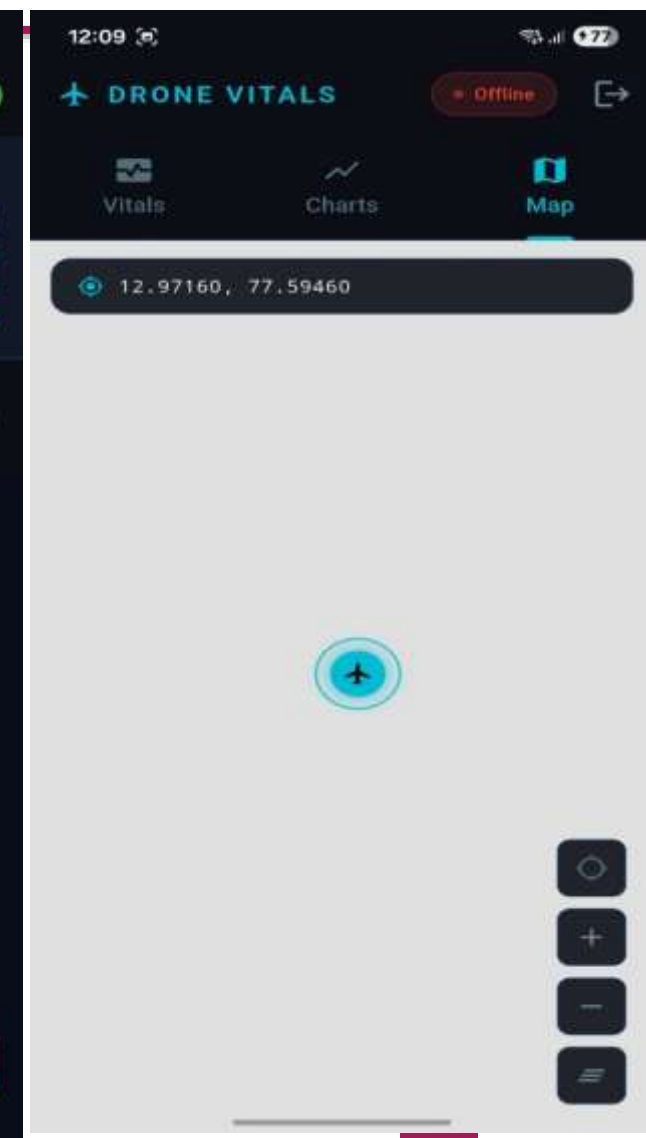
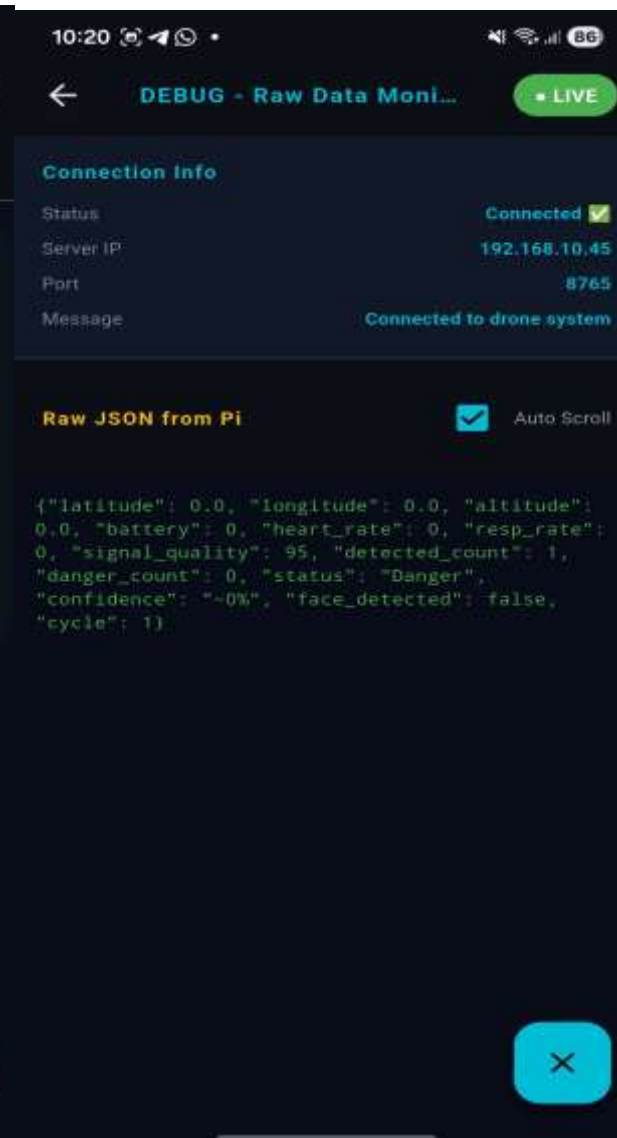
APP - Screens

Screen	What You See	How to Navigate
Connect Screen	Enter Pi IP (192.168.4.1) and port (8765). Tap CONNECT or Demo Mode.	First screen on launch.
Vitals Tab	4 cards: SpO2, Heart Rate, Resp.Rate, Signal Quality. Battery %, Altitude, GPS coords. Red alert banner if critical.	Default tab after connect.
Charts Tab	3 live scrolling line charts: SpO2 (last 60 readings) Heart Rate Respiratory Rate	Tap Charts tab.
Map Tab	Dark OpenStreetMap with live drone marker. Blue flight path trail. Follow/Zoom/Clear buttons.	Tap Map tab.

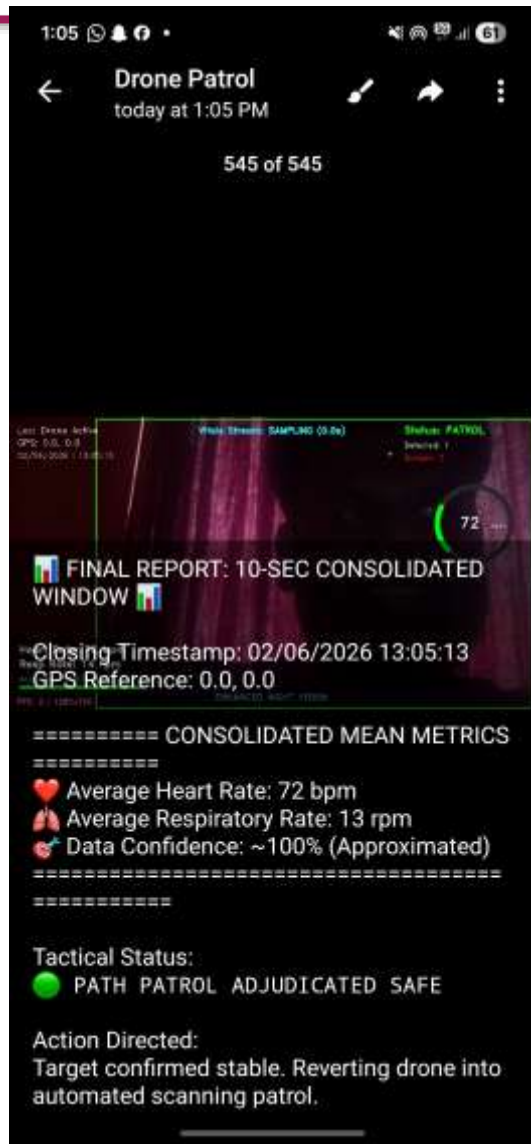
HEALTH THRESHOLDS

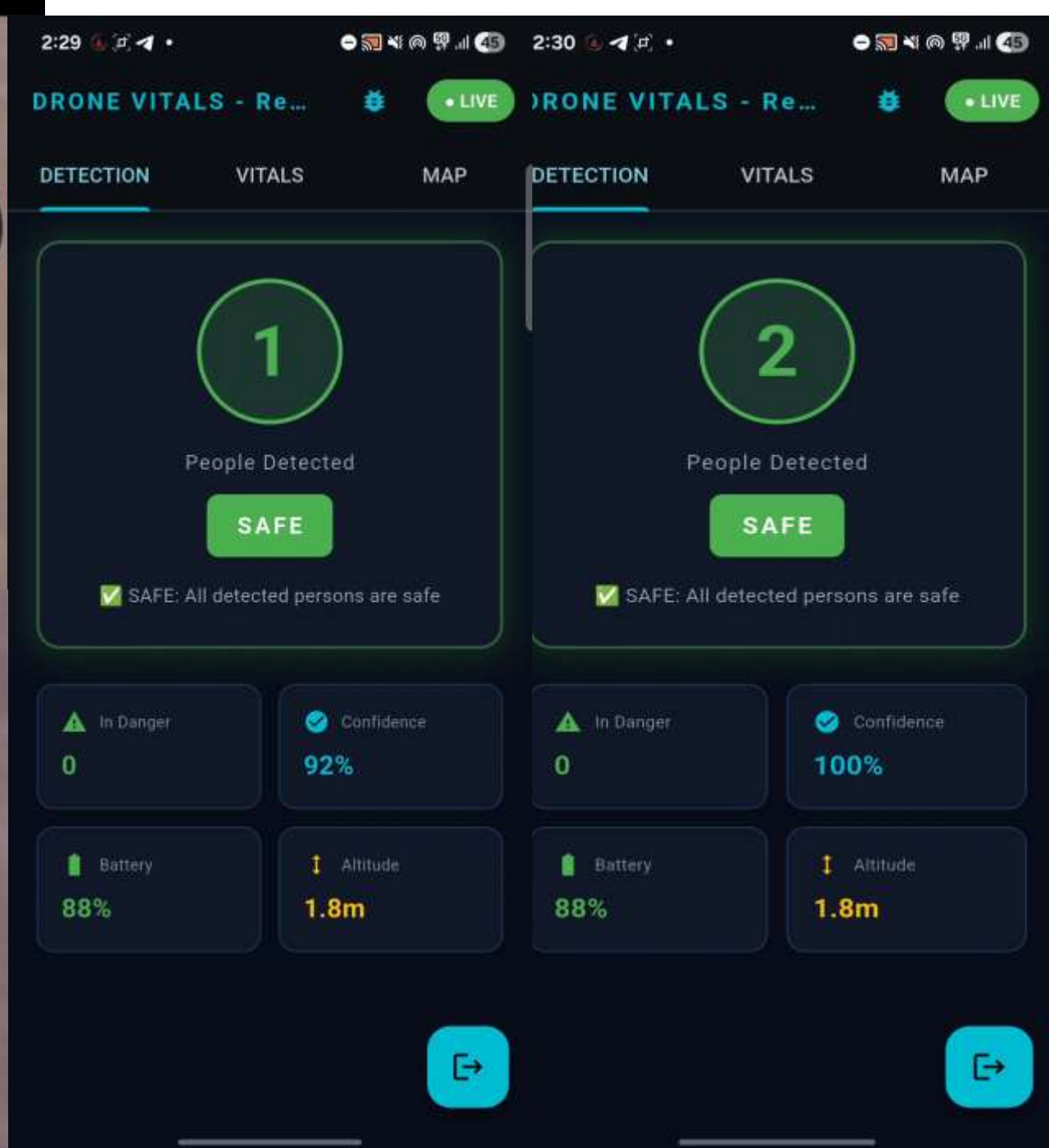
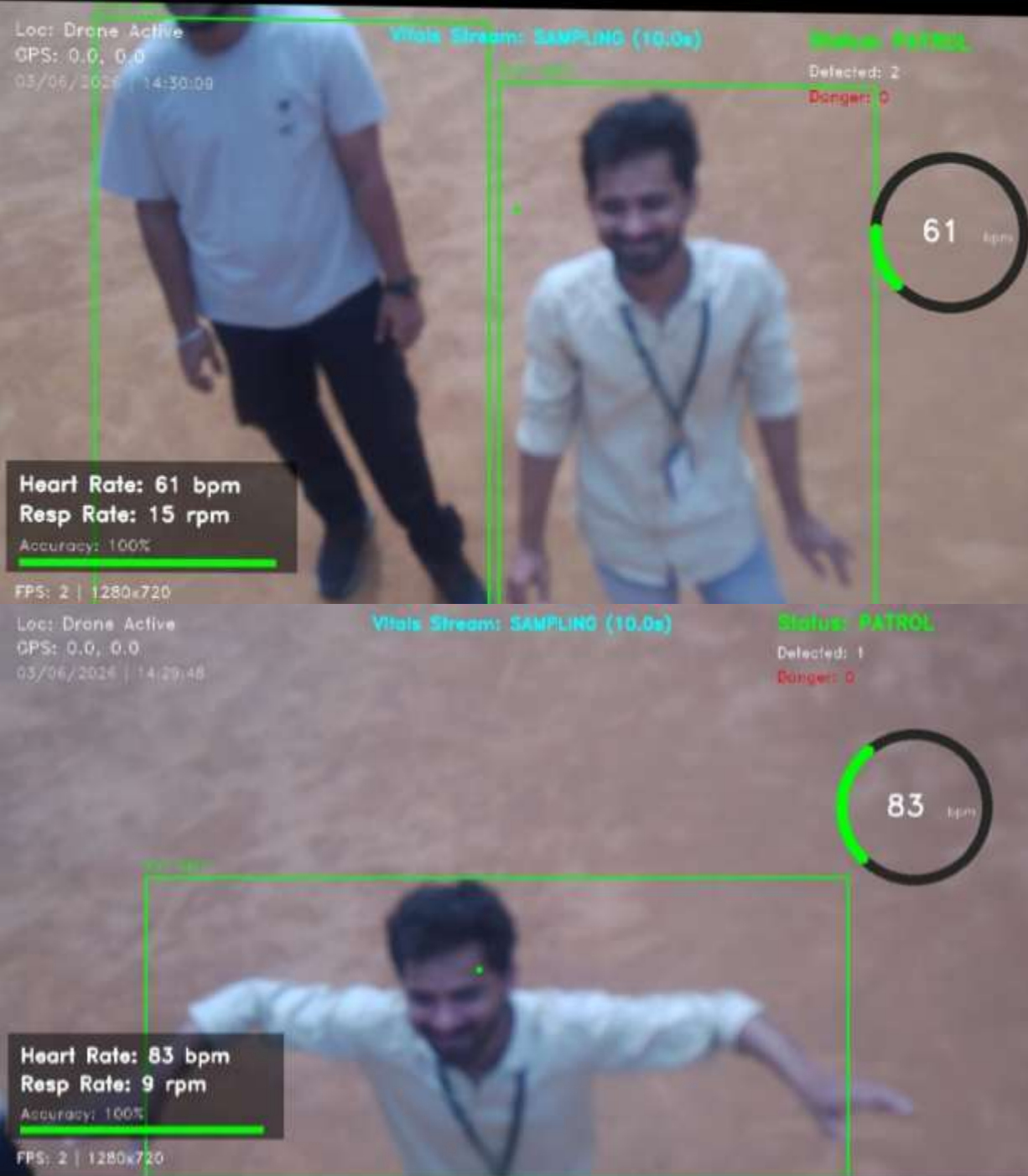
Vital Sign	Normal Range	Low (Warning)	High (Warning)	Critical (Alert)
Heart Rate	60 - 100 BPM	< 60 BPM	> 100 BPM	< 50 or > 120 BPM
Respiratory Rate	12 - 20 /min	< 12 /min	> 20 /min	< 8 or > 30 /min
Signal Quality	> 70%	50% - 70%	—	< 50%
Battery	> 30%	20% - 30%	—	< 20%

App Overview



Telegram Bot Output









6:30

76

Drone Patrol FINAL REPORT: 10-...



Vitals



Charts



Map

SpO₂ (Oxygen Saturation)

0.0 %

100
98
96
94
92
90
88
86
84

Heart Rate

145.0 BPM

160
150
140
130
120
110
100
90
80
70
60
50
40

Respiratory Rate

0.0 /min

30
28
26
24
22
20
18
16
14
12
10
8
6
4
2
0

Gantt Chart

Project Work - 2 16 weeks



Conclusion

- The developed system successfully integrates autonomous navigation, human detection, victim tracking, and contactless heart-rate monitoring into a single UAV platform.
- Using Computer Vision, AI, and rPPG techniques, the drone can detect a human, autonomously approach the target, and monitor vital signs without physical contact.
- The integration of Raspberry Pi 5, Pixhawk, MAVLink, and Android application enables real-time monitoring, target tracking, and information sharing with rescue teams.
- This system has the potential to improve search-and-rescue operations by providing rapid victim localization, preliminary health assessment, and efficient emergency response.

Sustainable Development Goals (SDG) Alignment

This project leverages Edge AI and autonomous robotics to contribute to three primary UN Sustainable Development Goals:

SDG 3: Good Health and Well-being
Contactless Triage: Utilizes rPPG and image processing to monitor heart and respiratory rates without physical contact, ensuring safe medical assessment in hazardous zones. **Emergency Aid:** Features a servo-actuated system to deliver medical kits and food packets directly to victims when abnormal vitals are detected.

SDG 9: Industry, Innovation, and Infrastructure
Edge Computing Excellence: Implements complex DSP math (FFT, IFFT) and AI models on constrained Raspberry Pi hardware using a resource-aware 3-Phase State Machine. **Autonomous Navigation:** Bridges Edge AI with the Pixhawk controller for GPS-denied navigation, representing a leap in resilient search-and-rescue (SAR) infrastructure.

SDG 11: Sustainable Cities and Communities
Disaster Resilience: Enhances the disaster management capabilities of urban areas by automating the localization of victims in collapsed or flooded structures. **Human-Centric SAR:** Uses multilingual support (English, Hindi, Kannada) to provide psychological comfort and instructions, reducing panic in high-stress disaster scenarios.

References

1. Marinopoulou, T., Lalas, A., Votis, K., & Tzovaras, D. (2023). *An AI-powered Acoustic Detection System Based on YAMNet for UAVs in Search and Rescue Operations*. *Inter-Noise 2023*.
2. Gemmeke, J. F., et al. (2017). *Audio Set: An ontology and human-labeled dataset for audio events*. *IEEE ICASSP*
3. Mösch L, Barz I, Müller A, Pereira CB, Moormann D, Czaplik M, Follmann A. *For Heart Rate Assessments from Drone Footage in Disaster Scenarios*. *Bioengineering (Basel)*. 2023 Mar 7;10(3):336. doi: 10.3390/bioengineering10030336. PMID: 36978727; PMCID: PMC10045207.
4. Capraro GA, Balmaekers B, den Brinker AC, Rocque M, DePina Y, Schiavo MW, Brennan K, Kobayashi L. *Contactless Vital Signs Acquisition Using Video Photoplethysmography, Motion Analysis and Passive Infrared Thermography Devices During Emergency Department Walk-In Triage in Pandemic Conditions*. *J Emerg Med*. 2022 Jul;63(1):115-129. doi: 10.1016/j.jemermed.2022.06.001. Epub 2022 Aug 6. PMID: 35940984.

Reference

- Knapp, C., & Carter, G. (1976). *The generalized correlation method for estimation of time delay*. IEEE Transactions on Acoustics, Speech, and Signal Processing.
- Valin, J.-M., Michaud, F., & Rouat, J. (2007). *Robust Sound Source Localization Using a Microphone Array on a Mobile Robot*. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
- Schröter, H., Rosenkranz, A., & Maier, A. (2022). *DeepFilterNet: A Low Complexity Speech Enhancement Framework for Full-Band Audio based on Deep Filtering*.

THANK YOU